

Number Systems.

1. Complete the table with equivalents Binary, Octal and hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), 1F.C (hex)

Decimal	Binary	Octal	hexa.
143.75	010001111.11	217.6	8F.C
27.625	11011.101	33.5	1B.A
299	100101011	453	12B
31.75	00011111.100	37.60	1F.C

2. Find equivalents

a) $324_5 = ?_{10}$

~~$3 \times 8^2 + 2 \times 8$~~
 $3 \times 5^2 + 2 \times 5 + 4 = \underline{\underline{(89)_{10}}}$

b) $32.23_{(4)} = ?_{(10)}$

$(3 \times 4) + (2) + \frac{2}{4} + \frac{3}{16}$
 $= 14 + 0.5 + 0.1875$
 $= \underline{\underline{14.6875_{(10)}}}$

c) $231_{10} = ?_7$

$$\begin{array}{r|l} 7 & 231-0 \\ 7 & 33-5 \\ \hline & 4 \end{array} \uparrow = \underline{\underline{450_{(7)}}}$$

d) $(189)_{10} = ?_6$

$$\begin{array}{r|l} 6 & 189-3 \\ 6 & 31-1 \\ \hline & 5 \end{array} = \underline{\underline{(513)_6}}$$

3. Find the Radix.

a) $\left(\frac{312}{20}\right)_r = (13.1)_r$

$$\frac{(312)_r}{(20)_r} = (13.1)_r \Rightarrow \frac{3r^2 + r + 2}{2r + 0} = r + 3 + \frac{1}{r}$$

$$3r^2 + r + 2 = 2r^2 + 6r + 2$$

$$x^2 - 5x = 0$$

$$x(x-5) = 0$$

$$\underline{\underline{x=5}}$$

$$b) 264x + 136x = 433x$$

$$2x^4 6x + 4 + x^2 + 3x + 6 = 4x^2 + 3x + 3$$

$$3x^2 + 9x + 10 = 4x^2 - 3x - 3 = 0$$

$$-x^2 + 6x + 7 = 0$$

$$x^2 - 6x - 7 = 0$$

$$(x-7)(x+1) = 0$$

$$\underline{\underline{x=7}}$$

$$c) 221505_8 = 12345_x$$

$$(2 \times 8^5) + (2 \times 8^4) + (2 \times 8^3) + (5 \times 8^2) + 5 = x^4 + (2 \times x^3) + (3 \times x^2) + (4 \times x) + 5$$

$$74565 = x^4 + 2x^3 + 3x^2 + 4x + 5$$

By solving x=16

4. Addition.

$$a) 1101_2 + 11011111_2 =$$

$$\begin{array}{r} 11011111 \\ 00001101 \\ \hline 11101100_2 \end{array}$$

$$b) 77652_8 + 76543_8$$

$$\begin{array}{r} 77652 \\ + 76543 \\ \hline 176415 \end{array}$$

$$c) 578A_{16} + ABDE_{16}.$$

$$\begin{array}{r} 578A \\ ABDE \\ \hline 10368 \end{array}$$

$$d) 234_6 + 315_6.$$

$$\begin{array}{r} 234 \\ 315 \\ \hline 353 \end{array}$$

$9_{(10)} \rightarrow (27)_6$

5. Binary $\rightarrow$ Gray code.

a) 10101_2

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$10101_2 \rightarrow \underline{\underline{11111}}_{\text{gray}}$$

b) 01110_2

$$0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

$$1 \oplus 1 = 0$$

$$1 \oplus 0 = 1$$

$$01110_2 \rightarrow \underline{\underline{01001}}_{\text{gray}}$$

6. Gray $\rightarrow$ Binary.

a) 01101_{gray}

Taking Xor

$$0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

$$01101_{\text{gray}} \rightarrow \underline{\underline{01001}}_2$$

b) 10101_{gray}

$$1 \oplus 0 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

$$\underline{\underline{11001}}_2$$